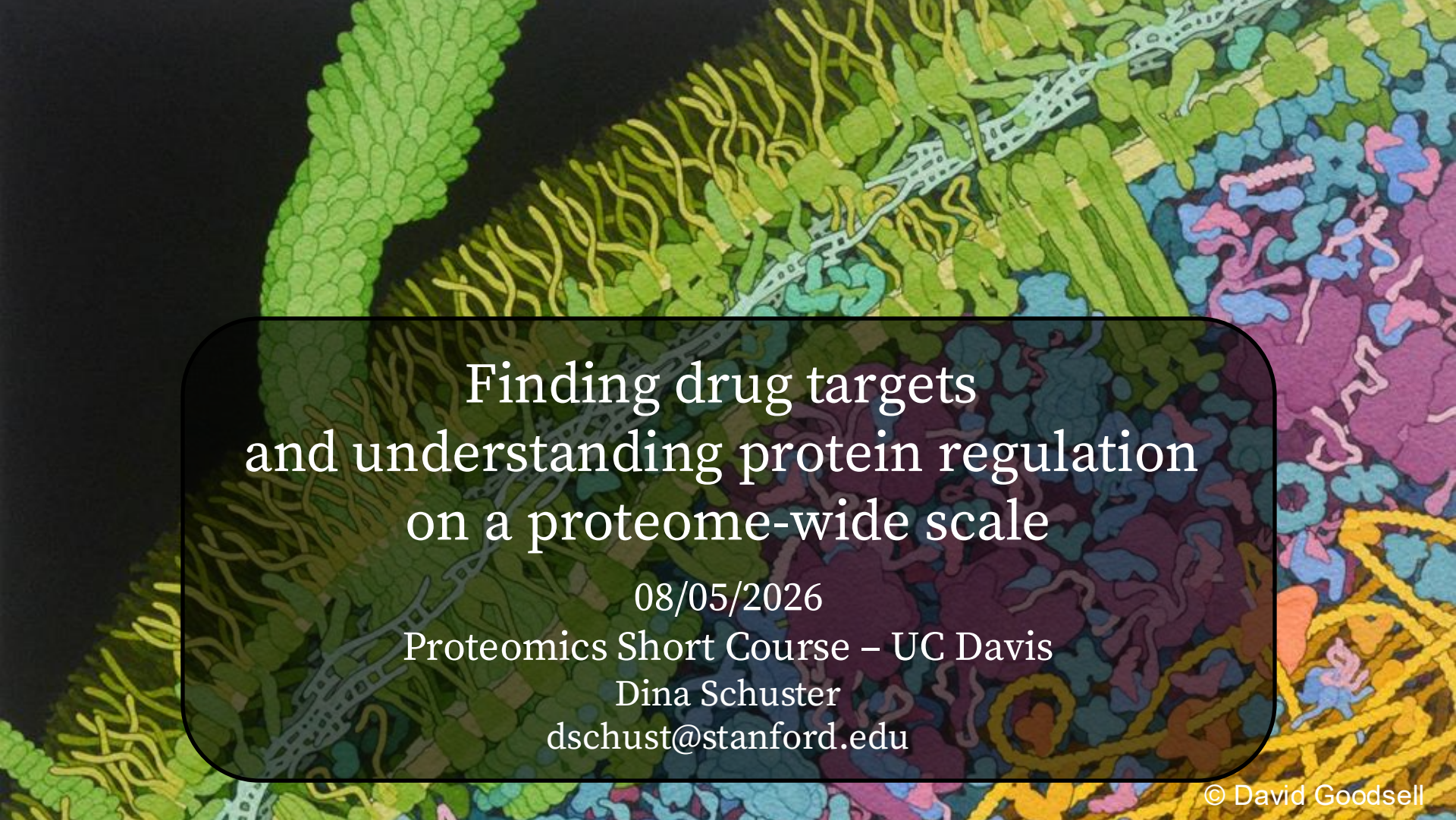




Finding drug targets and understanding protein regulation on a proteome-wide scale

08/05/2026

Proteomics Short Course – UC Davis

Dina Schuster

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Personal background

- BSc and MSc in (Analytical) Chemistry
University of Vienna, Austria
(Gunda Koellensperger, Christopher Gerner)
- PhD in Biochemistry
ETH Zurich, Switzerland
(Volodymyr Korkhov, Paola Picotti)
- PostDoc
Utrecht University, Netherlands
(Albert Heck)
- Since 2025: Assistant Director, Chemoproteomics
Nucleus, Sarafan ChEM-H, Stanford University



Nucleus

The Nucleus at Sarafan ChEM-H is a group of breakthrough technology labs supercharging research on the path from molecules to medicines



chemh.stanford.edu/research/nucleus

- At Stanford University (ChEM-H), formerly called “Knowledge Centers”
- A collection of expert labs working in different fields of research (Medicinal Chemistry, Metabolomics, Macromolecular Structure, High Throughput Screening, Cryo-EM, **(Chemo-)Proteomics**)
- Provide access to high-end research equipment
- Train researchers in instrument use and methodologies
- Support research and act as collaborators
- No “sample drop-off” service center

The (Chemo-)proteomics group



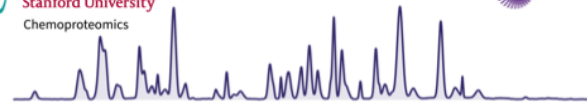
Dina Schuster
Assistant Director



TimmsTOF HT
aka “Timmy”
+ Evosep Eno
(or nanoElute 2)



Michelle Kinney
Research Assistant



Collaborate With The Chemoproteomics Group

Mass spectrometry-based protein analysis at the Nucleus, Sarafan ChEM-H
Cutting-edge instrumentation and support from experimental design to data interpretation.

Why proteomics?

Proteins are the molecular machines that carry out cellular functions. Proteomics allows us to study the behavior of thousands of proteins in a single experiment. This helps us better understand physiology in health and disease.

Study proteins and their interactions.

The Chemoproteomics group gives researchers access to a wide range of mass spectrometry-based methods for studying proteins and their interactions with other proteins or small molecules — across different sample types: from purified or enriched proteins to cell lysates, biofluids or tissue biopsies.

Hands-on support, not just instrument time.

Every project comes with one-on-one guidance: help with experimental design, sample preparation, LC-MS/MS acquisition, and data analysis — we want you to get meaningful, insightful results, not just data tables.

New to proteomics? No problem! We will teach you everything you need to know.

GLOBAL
PROTEOMICS

ABSOLUTE
PROTEIN
QUANTIFICATION

PTM
ANALYSIS

INTERACTION
MAPPING

STRUCTURAL
PROTEOMICS

INTERACTIVE
PROTEIN
ANALYSIS

& MORE

Curious? Contact us!

Dina Schuster, Assistant Director, Chemoproteomics
dschust@stanford.edu



Scan to visit us online
chemoproteomics.stanford.edu

The (Chemo-)proteomics group

- Provide help with bottom-up proteomics and native/intact MS measurements
 - Popular experiment types/projects:
 - Protein abundance changes (cell systems, tissue biopsies, biofluids)
 - Post-translational modifications
 - Protein-protein interactions
 - Protein-small molecule interactions
 - Structural proteomics
 - ... and more
- } Regulation of protein function/activity



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How can protein function be assessed?

Activity assays

Overexpression

Subcellular localization

Pull-down experiments

Knock-down experiments

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3D structure

Thermal stability

Protein abundance profiling

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Protein abundance profiling

**How can protein structural alterations be measured
on a proteome-wide scale?**

Mutations

Limited proteolysis

- **Brief proteolytic digestion** of proteins using various enzymes
- Cleavage of **accessible and flexible**, often disordered, regions
- Used to stabilize proteins for X-ray crystallography
- Used to identify protein structural changes and protein-protein interactions with SDS-PAGE
 - Gel bands were excised and sequenced using N-terminal sequencing by Edman degradation (Newton et al., 1984, etc.)
- Invention of electrospray ionization (ESI) mass spectrometry (Fenn et al. 1989) allows for mass determination of mixtures

Structural analysis of yeast calmodulin

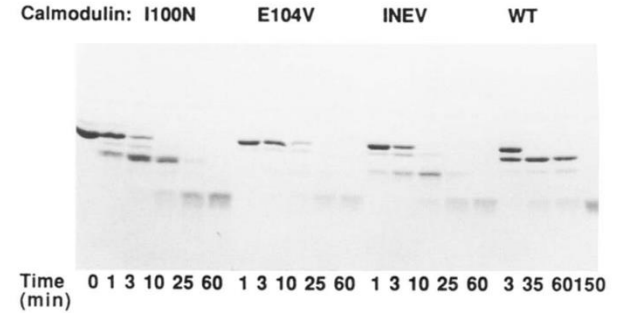


Fig. 3. Partial digests of wild-type and mutant forms of calmodulin performed in the presence of Ca^{2+} . The trypsin/calmodulin weight ratio was 1:300 except the 150-min wild-type time point, which contained a ratio of trypsin to calmodulin of 1:62.5. The experiment was conducted as described in the Materials and methods and the legend for Figure 2. Digestions of wild-type and mutant forms of calmodulin were done simultaneously. Each lane contained the products of 2.0 μg of calmodulin.

Brockerhoff et al., Protein Science, 1992

Limited proteolysis-coupled mass spectrometry

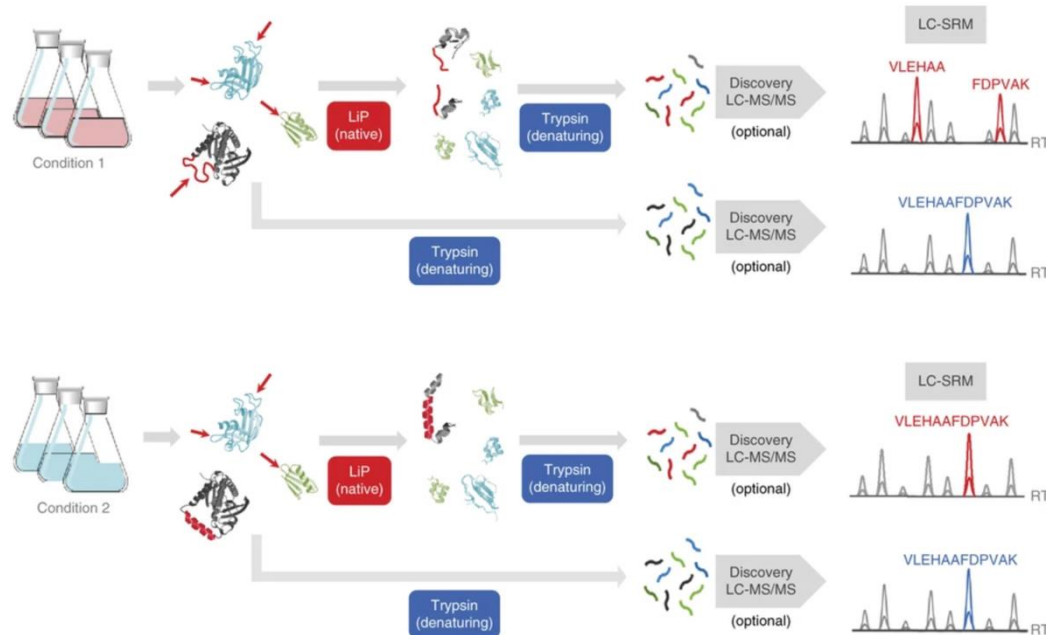
- First established in early 1990s
- Now combined with data independent acquisition for high proteome coverage
- Subsequently used to study
 - **Protein structures** in proteomes (Feng et al., 2014, Nature Biotechnology)
 - **Thermal unfolding** (Leuenberger et al., 2017, Science)
 - **Protein-metabolite interactions** (Piazza et al., 2018, Cell)
 - **Protein-drug interactions** (Piazza et al., 2020, Nature Communications)
 - **Protein functional alterations** (Cappelletti, Hauser, Piazza, et al., 2021, Cell)
 - **Membrane protein interactions & regulation** (Barret & Schuster, et al. 2023, PNAS)
 - **Protein-protein interactions** (Holfeld et al., 2024, Molecular systems biology)
 - **Protein-metal-binding** (Quast et al., 2025, bioRxiv)
 - **Structural alterations in living cells** (Elsaesser et al., 2026, Molecular systems biology)

Global analysis of protein structural changes in complex proteomes

Yuehan Feng, [Giorgia De Franceschi](#), [Abdullah Kahraman](#), [Martin Soste](#), [Andre Melnik](#), [Paul J Boersema](#), [Patrizia Polverino de Laureto](#), [Yaroslav Nikolaev](#), [Ana Paula Oliveira](#) & [Paola Picotti](#) 

Nature Biotechnology **32**, 1036–1044 (2014) | [Cite this article](#)

28k Accesses | 394 Citations | 108 Altmetric | [Metrics](#)



Peptide-level structural resolution

Sample preparation

- (incubation with interactor)
- Brief (1-5 min) digestion with unspecific protease (**proteinase K**)
- Quench digest
- Tryptic digest
- C18 clean-up

Global analysis of protein structural changes in complex proteomes

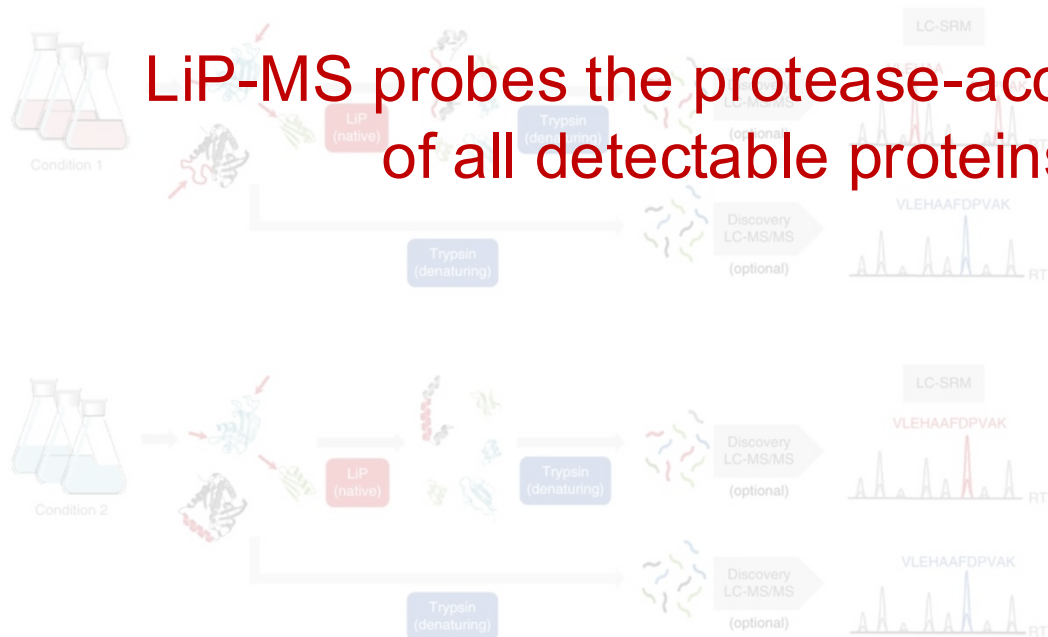
Yuehan Feng, [Giorgia De Franceschi](#), [Abdullah Kahraman](#), [Martin Soste](#), [Andre Melnik](#), [Paul J Boersema](#), [Patrizia Polverino de Laureto](#), [Yaroslav Nikolaev](#), [Ana Paula Oliveira](#) & [Paola Picotti](#) 

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Peptide-level structural resolution

LiP-MS probes the protease-accessible surface area of all detectable proteins in a lysate.



Sample preparation

• Brief (1-5 min) digestion with unspecific protease (proteinase K)

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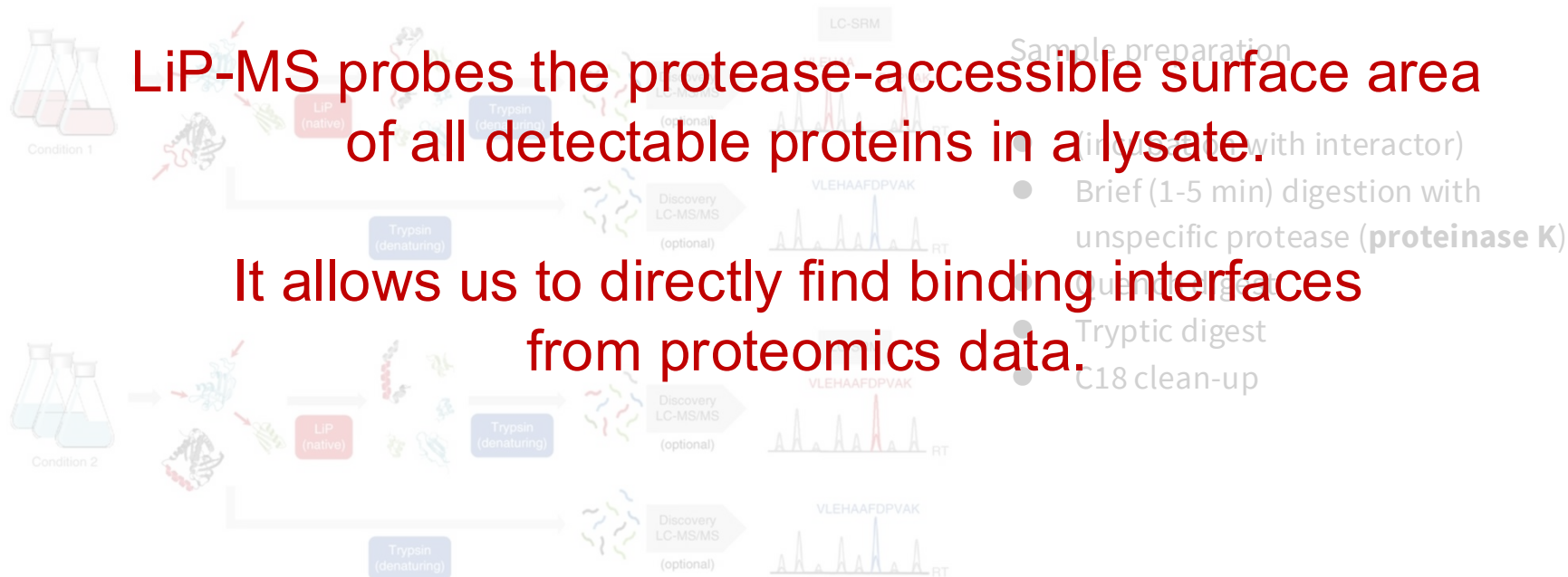
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Peptide-level structural resolution

LiP-MS probes the protease-accessible surface area of all detectable proteins in a lysate.

It allows us to directly find binding interfaces from proteomics data.



Dose-response assays

- “LiP-school” experiment: interaction of Rapamycin with its target FKBP1A
- Titration of lysate with increasing concentrations of rapamycin

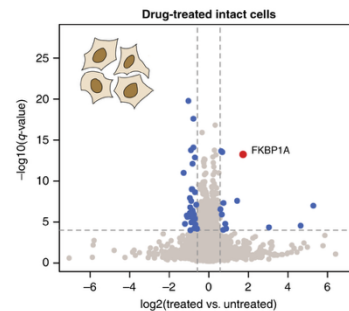
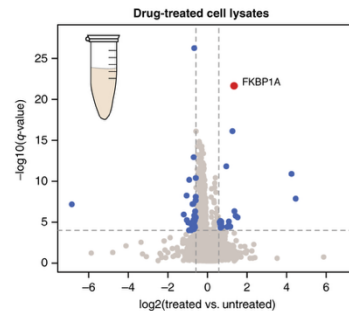


- Estimation of EC_{50}
- No estimations of ideal concentration needed
- Interactions/conformational changes are not always captured within fold change/p-value cutoffs



- 40-50 samples instead of 8 samples (4 replicates)

a



LiP-MS for drug target deconvolution

- **LiP-Quant:** a machine-learning pipeline built on LiP-MS to identify drug targets and approximate binding sites
- Works directly in cell lysates: no labeling, no prior target hypothesis needed
- Validated across diverse compound classes: kinase, phosphatase, and membrane-protein-targeted drugs

Article | [Open access](#) | Published: 21 August 2020

A machine learning-based chemoproteomic approach to identify drug targets and binding sites in complex proteomes

Ilaria Piazza, Nigel Beaton, Roland Bruderer, Thomas Knobloch, Crystel Barbisan, Lucie Chandat, Alexander Sudau, Isabella Siepe, Oliver Rinner, Natalie de Souza, Paola Picotti & Lukas Reiter

[Nature Communications](#) 11, Article number: 4200 (2020) | [Cite this article](#)

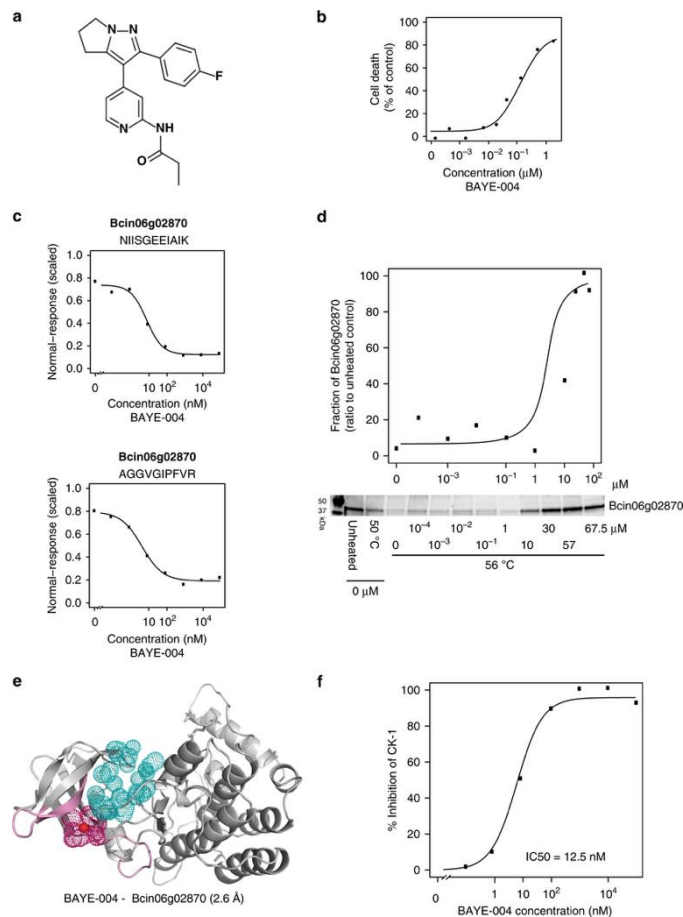


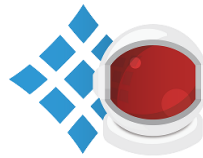
Fig. 4: Identification of the kinase target and its binding site of a fungicide.

Data acquisition

- Prepare a pool of LiP and TC control samples
 - Optional: fractionate with high pH reversed phase fractionation
- Generate data-dependent acquisition (DDA) library of pooled control samples
- Measure all other samples with label-free data independent acquisition (DIA)
- Variable width-window strategy (depends on instrument)
- **Library-free DIA also works well!**

Window	Mass-to-charge ratio (m/z)	Charge (z)	Isolation window (m/z)
1	358	2	16
2	373	2	16
3	388	2	16
4	403	2	16
5	418	2	16
6	433	2	16
7	448	2	16
8	463	2	16
9	478	2	16
10	493	2	16
11	508	2	16
12	523	2	16
13	538	2	16
14	553	2	16
15	568	2	16
16	583	2	16
17	598	2	16
18	613	2	16
19	628	2	16
20	643	2	16
21	659	2	18
22	676	2	18
23	693	2	18
24	710	2	18
25	727	2	18
26	744	2	18
27	761	2	18
28	778	2	18
29	795	2	18
30	813	2	20
31	832	2	20
32	851	2	20
33	870	2	20
34	889	2	20
35	908	2	20
36	929.5	2	25
37	953.5	2	25
38	977.5	2	25
39	1006.5	2	35
40	1048	2	50
41	1111	2	78

Appendix Table S1: Definition of variable windows for DIA-MS measurements.



Data analysis

- Generate DDA library in Spectronaut / or run directDIA search (or other software)
 - Search against (human) proteome
 - Include common contaminants (keratins, digestion enzymes, cell culture contaminants, etc.)
 - Digest type: **semi-specific**
- Search DIA files using generated spectral library
- Check data quality to refine cutoffs if necessary

Enzymes / Cleavage Rules

☒ Trypsin/P

☐ Trypsin

☒ LysC

☐ LysC/P

☐ GluC

☐ AspN

☐ ArgC

Semi-specific

KR

52

5

2

☒

Digest Type

Decoy Generation Rule

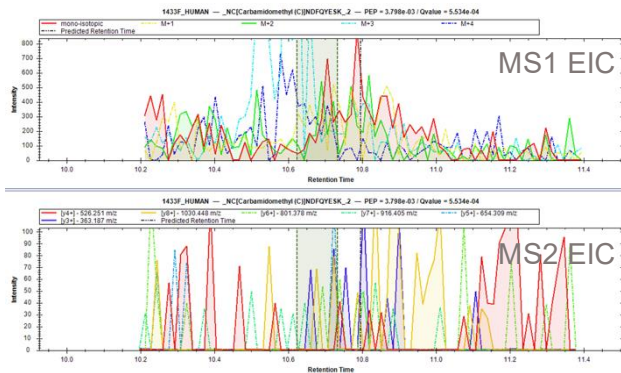
Max Peptide Length

Min Peptide Length

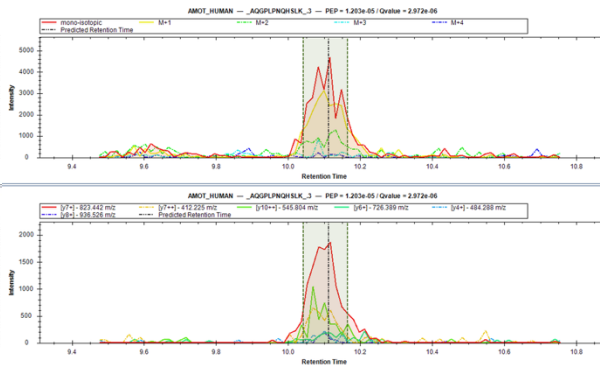
Missed Cleavages

Toggle N-terminal M

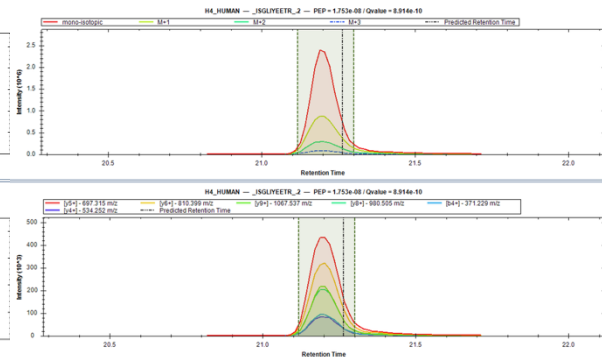
Bad ID



OK ID



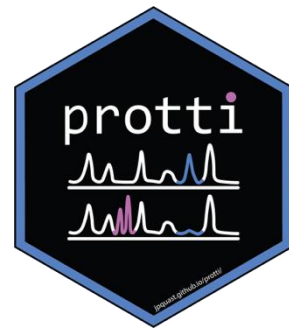
Great ID



Protti

(R package for bottom-up proteomics & LiP-MS data analysis)

- Export (peptide) data from Spectronaut (or other search software)
- Load necessary R packages
- Check data quality, filter if necessary, normalize data, impute if necessary; for cell treatment (instead of lysate) assays: normalize for protein abundance changes
- Perform statistical hypothesis testing (Welch's t-test, limma moderated t-test, proDA)
- Fit dose-response curves (drc package)
- Other functions: GO-enrichment, fetch information from online databases (e.g. UniProt, STRING, ChEBI, and more), download 3D structures, map hits onto protein structures, etc.



Quast & Schuster, et al., Bioinformatics Advances, 2022

Dose-Response Data Analysis Workflow

Jan-Philipp Quast

2026-01-14

Source: vignettes/data_analysis_dose_response_workflow.Rmd

Introduction

This vignette will take you through all important steps for the analysis of dose-response experiments with **protti**. For experiments with treatment titrations (i.e. samples treated with different amounts of the same compound, protein, metabolite, element, ion, etc.) you can fit sigmoidal log-logistic regression curves to the data and determine significantly changing precursors, peptides or proteins based on Pearson correlation and ANOVA q-value.

data analysis tutorials and scripts on website



How I use AI for data analysis

- Make sure you know how the AI will use your data (Stanford has an Enterprise subscription for Claude (data stays within Stanford environment))
- Use tools that are well documented and truste
- Give specific prompts and make sure you defin your data format (e.g. paste first line of input)
- Ask AI to write code that you can understand and proof-read if necessary

Claude example

Please analyze the data in the
the following conditions:

folder. This is proteomics data with

* WT eGFP

*
*
*
*

Please compare the KO to the WT using R. Write an .rmd script using tidy language and protti. Use proDA for the statistical hypothesis test and highlight the proteins that were knocked out. Make sure that the plots are interactive, add QCs and for the volcano add a label with long protein names.

Also export this as an .html, make sure that the figures knit nicely without any weird size errors.

📄 ↺ 🔗 6 days ago

Alternative tools for data analysis

- **MSStatsLiP** (Kohler et al., 2025)
 - Robust data analysis pipeline based on MSStats
 - Ideal for data sets with protein abundance changes
- **FLiPPR** (Manriquez-Sandoval et al., J Proteome Res, 2024)
 - FragPipe LiP Processor
- **LiPAnalyzerR** (Nagel et al., MCP, 2025)
 - Pipeline that applies regression models to identify structural changes while removing unwanted variation (from e.g. alternative splicing or PTMs)
- **ProteoMeter** (Rozum et al., bioRxiv, 2025)
 - Integration of LiP-MS data with PTM and XL-MS data

An alternative to LiP-MS: PELSA

- Newer, simplified version of LiP-MS
- Uses excess of trypsin instead of proteinase K
- Lower spatial resolution, much simpler protocol, better for high-throughput implementations

Brief Communication | Published: 10 December 2024

A peptide-centric local stability assay enables proteome-scale identification of the protein targets and binding regions of diverse ligands

[Kejia Li](#), [Shijie Chen](#), [Keyun Wang](#), [Yan Wang](#), [Lianji Xue](#), [Yuying Ye](#), [Zheng Fang](#), [Jiawen Lyu](#), [Haiyang Zhu](#), [Yanan Li](#), [Ting Yu](#), [Feng Yang](#), [Xiaolei Zhang](#), [Siqi Guo](#), [Chengfei Ruan](#), [Jiahua Zhou](#), [Qi Wang](#), [Mingming Dong](#), [Cheng Luo](#) ✉ & [Mingliang Ye](#) ✉

Nature Methods **22**, 278–282 (2025) | [Cite this article](#)

20k Accesses | 50 Citations | 43 Altmetric | [Metrics](#)

Technical Report | [Open access](#) | Published: 05 November 2025

High-throughput peptide-centric local stability assay extends protein–ligand identification to membrane proteins, tissues and bacteria

[Kejia Li](#), [Clement M. Potel](#), [Isabelle Becher](#), [Nico Hüttmann](#), [Martin Garrido-Rodriguez](#), [Jennifer Schwarz](#), [Mira Lea Burtscher](#) & [Mikhail M. Savitski](#) ✉

Nature Structural & Molecular Biology **33**, 184–192 (2026) | [Cite this article](#)

16k Accesses | 3 Citations | 66 Altmetric | [Metrics](#)



Summary

- LiP-MS & PELSA are tools to study proteome-wide structural changes
 - Can be used to map binding sites of proteins, small molecules, metabolites, etc.
- Peptide-level resolution of binding events or structural alterations directly *in situ*
- Allows for unbiased identification of interaction partners
- Limitation(s):
 - LiP-MS throughput is limited (rarely combined with TMT (Koudelka et al., 2025))
 - Structural alterations only detected if protein region is covered → **limited to higher abundance proteins**

Activity assays

Overexpression

Subcellular localization

Pull-down experiments

Knock-down experiments

How can protein function be assessed?

3D structure

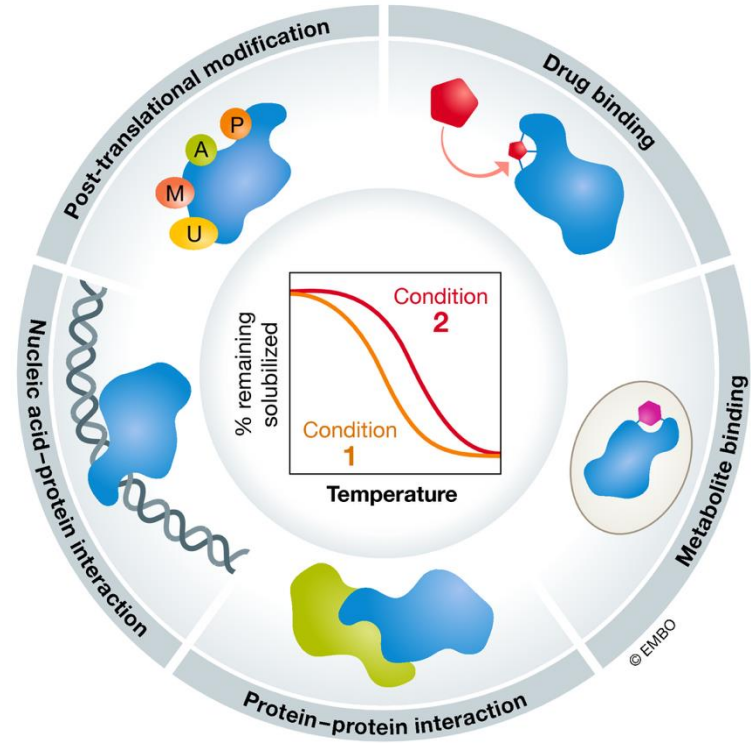
Thermal stability

Protein abundance profiling

Mutations

Thermal proteome profiling (TPP)

- Proteins subjected to heat denature, unfold, and become insoluble
- Combines CETSA (cellular thermal shift assay) with quantitative mass spectrometry
- When proteins interact with a compound - or any other interaction partner - their melting behavior changes



Mateus et al., Molecular Systems Biology, 2020

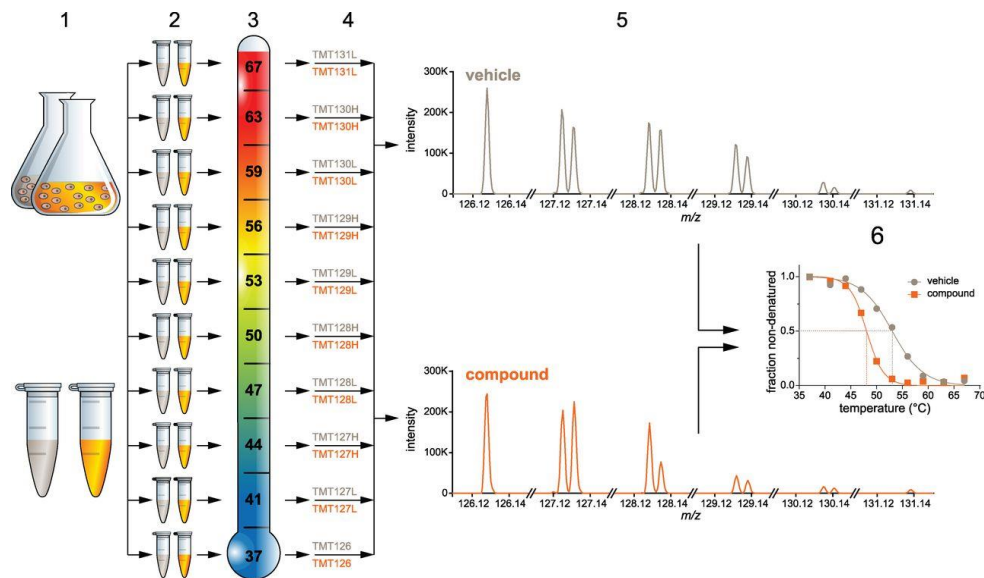
How to TPP

- Split lysate or intact cells into aliquots across a temperature gradient (e.g. 37–67 °C, 8-10 points)
- Treat aliquots with/without compound or other interactor before heating
- Brief heat pulse (~3 min per temperature), then cool to room temperature
- Lyse (if intact cells) and ultracentrifuge or filter to remove denatured/aggregated protein - keep the soluble fraction
- Digest soluble protein (label peptides per temperature with TMT for multiplexing)
- Pool samples, then LC-MS/MS (Orbitrap needed to resolve closely-spaced TMT reporter ions)
- **Fit melting curves per protein across temperatures; compare T_m shifts between conditions to find interactors**

Savitski et al., Science, 2014; Franken et al., Nature Protocols, 2015

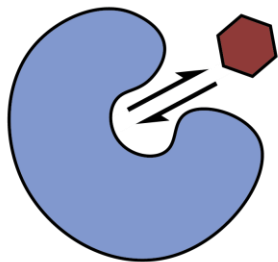
Target deconvolution with TPP

- Live human cells treated with kinase inhibitors, then thermally profiled by TPP
- Captured both direct targets and downstream pathway effects (e.g. dasatinib shifted BCR-ABL pathway proteins CRK/CRKL)
- **Revealed an unexpected off-target explaining a clinical side effect**



Covalent vs. non-covalent interactions

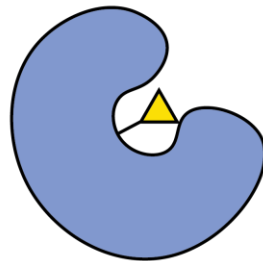
Non-covalent



Reversible binding
(small molecule, metabolite, protein)

Can be detected via LiP-MS/PELSA/TPP

Covalent



Direct bond between small molecule and protein
(reactive residues: Cys, Ser, Lys, ...)

Detection via covalent probe (ABPP)

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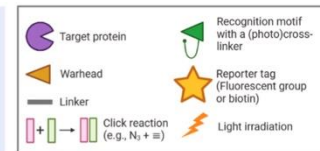
Mutations

Activity-based protein profiling

- Invented > 25 years ago (Bogyo/Cravatt)
- Small-molecule activity-based probes (ABPs) **covalently** react with the active site of enzymes
 - Probe = reactive warhead + linker + reporter
 - (biotin, fluorophore, or clickable handle)
- **Labels the catalytically active fraction of an enzyme** functional state, not just abundance
- Enrich labeled proteins (e.g. biotin-streptavidin), digest, then LC-MS/MS
- **Applications:** covalent drug target ID, selectivity profiling, discovering novel active enzymes

The Star of this tool: The Probe

The key to ABPP methodology is the design of the probe, which are small molecules built to covalently bind to the active site of a certain protein target(s).

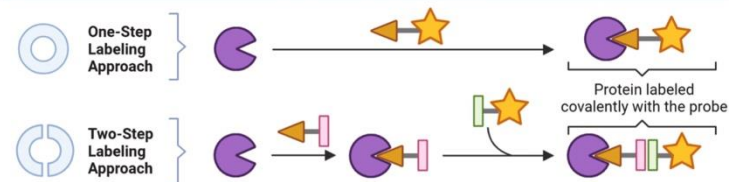


Warhead: Usually, is an electrophilic group that can covalently bind a conserved active site containing a nucleophilic residue

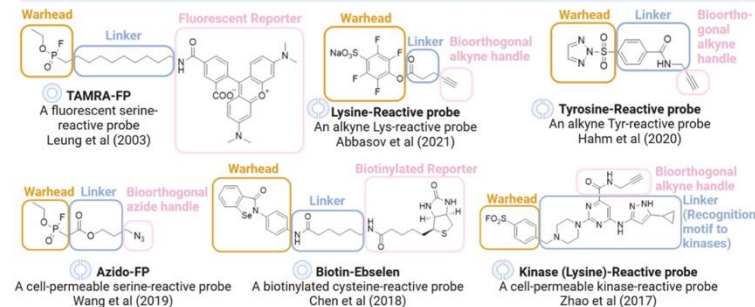
Reporter tag: It could be a fluorescent group, a biotinylated moiety, or a biorthogonal group (azide N_3 or alkyne $\equiv$)

Linker: It has 3 functions; modulate reactivity, improve selectivity, and provide the spacer between a tag and a warhead

Active-based probes (ABPs)



Examples reported in literature

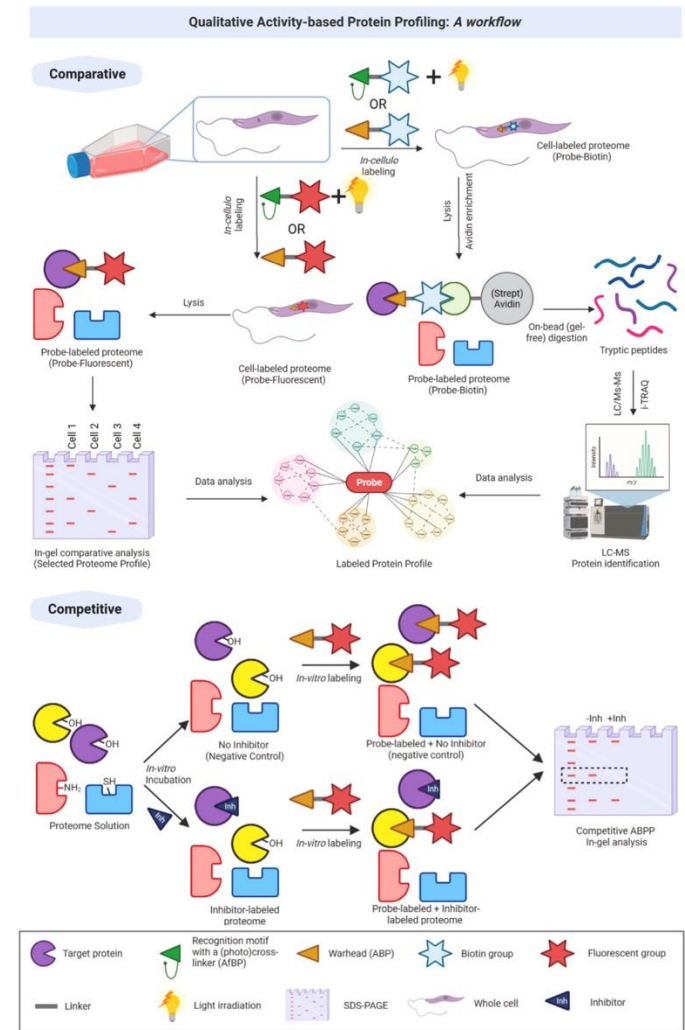


Porta & Steel, 2023



How to ABPP

- Treat live cells or lysate with an activity-based probe (ABP)
- Probe reacts covalently with the active site of catalytically active enzymes only
- Click chemistry to attach biotin/fluorophore tag (if using a minimal clickable probe)
- Enrich tagged proteins (e.g. streptavidin beads), wash, then digest with trypsin
- **LC-MS/MS to identify and quantify probe-modified sites across the proteome**



Summary

Three complementary, proteome-wide approaches to studying protein function without genetic perturbation:

	LiP-MS / PELSA	TPP	ABPP
Readout	Accessibility change / Structural change	Thermal stability shift	Catalytic activity / catalytic residues
Interaction type	Non-covalent (indirect)	Non-covalent (indirect)	Covalent (direct)
Best for	Mapping binding interfaces at peptide resolution no chemistry needed	Drug target ID/engagement, no chemistry needed	Selectivity profiling of covalent probes/drugs



Thank you!